

Question 17**1 mark**

How many different arrangements are possible when arranging all of the letters of the word SEPTEMBER?

a) $9!$

b) $6!3!$

c) $\frac{9!}{3!}$

d) $\frac{6!}{3!}$

Question 18**1 mark**

Which one of the following angles terminates in Quadrant III?

a) 3 radians

b) $\frac{7\pi}{5}$ radians

c) -210°

d) 500°

Question 19**1 mark**

There are 13 terms in the expansion of $(3x - y)^{2n}$. Determine the value of n .

a) 6

b) 6.5

c) 7

d) 26

Question 20**1 mark**

Which of the following is true about the periods of the three functions below?

$$f(\theta) = 2\sin 3\left(\theta - \frac{\pi}{2}\right)$$

$$g(\theta) = \sin 3\theta + 6$$

$$k(\theta) = 3\sin \theta + 6$$

- a) The graphs of $f(\theta)$ and $g(\theta)$ have the same period.
- b) The graphs of $g(\theta)$ and $k(\theta)$ have the same period.
- c) All of the graphs have the same period.
- d) None of the graphs have the same period.

Question 21**1 mark**

Which of the following represents the general solution to the equation $\tan \theta = -1$?

- a) $\theta = \frac{\pi}{4} + 2k\pi, k \in \mathbb{I}$
- b) $\theta = \frac{\pi}{4} + k\pi, k \in \mathbb{I}$
- c) $\theta = \frac{3\pi}{4} + 2k\pi, k \in \mathbb{I}$
- d) $\theta = \frac{3\pi}{4} + k\pi, k \in \mathbb{I}$

Question 22**1 mark**

If $(3, -2)$ is a point on the graph of $y = f(x)$, what point must be on the graph of $y = 2f(x+1)$?

- a) $(4, -1)$
- b) $(4, -4)$
- c) $(2, 1)$
- d) $(2, -4)$

Question 23

1 mark

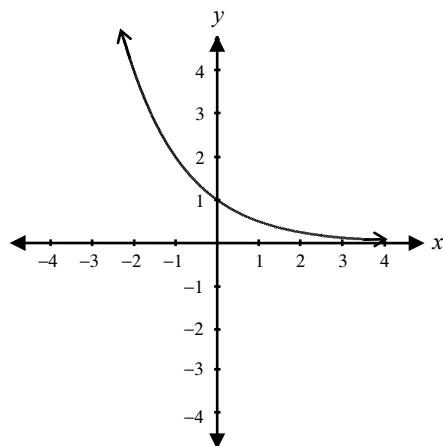
Which equation is represented by the graph sketched below?

a) $y = \left(\frac{1}{2}\right)^{-x}$

b) $y = \left(\frac{1}{2}\right)^x$

c) $y = 2^x$

d) $y = -2^x$



Question 24

1 mark

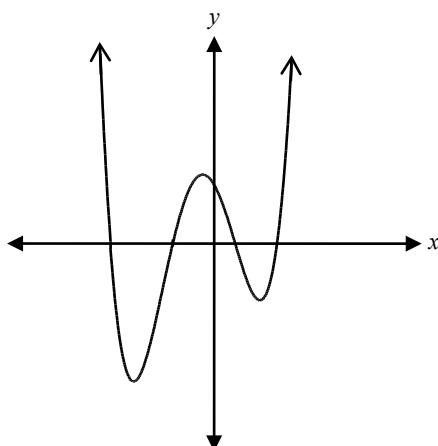
What is the degree of the polynomial represented below?

a) 2

b) 3

c) 4

d) 5

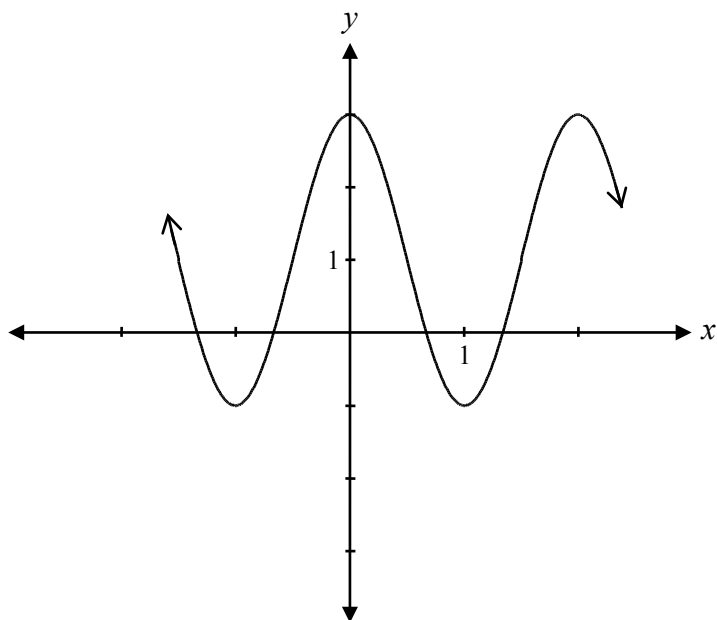


Question 25

1 mark

117

Given the graph of $y = 2 \cos \pi x + 1$ below, determine another equation that will produce the same graph.



$y =$ _____

Question 26

2 marks

118

Given $f(x) = 3$ and $g(x) = x + 2$, determine the domain and range of $h(x) = \frac{f(x)}{g(x)}$.

Domain: _____

Range: _____

Question 27

2 marks

119

Explain how to find the exact value of $\sec\left(\frac{19\pi}{6}\right)$.

Question 28

1 mark

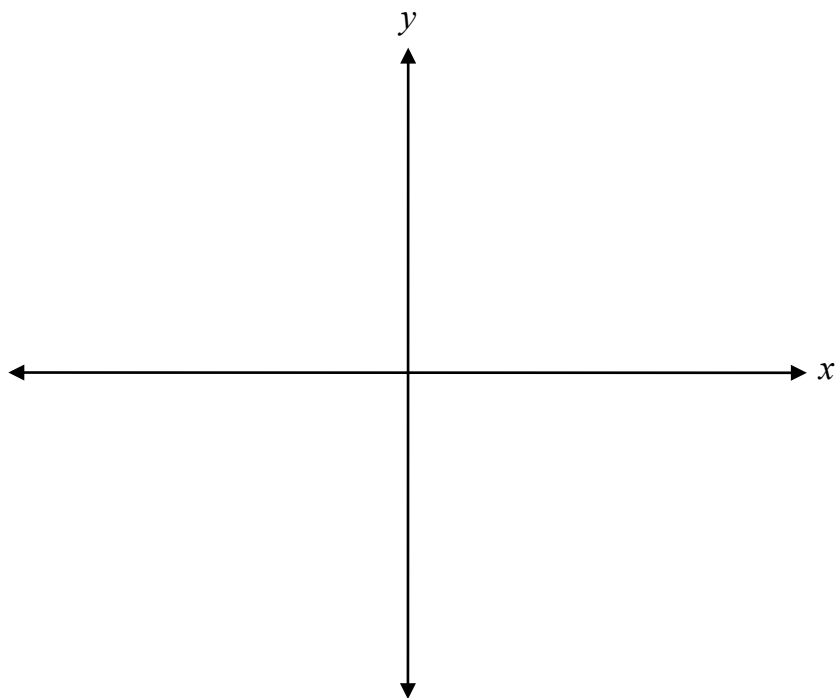
120

Given $f(x) = 4 - x$, verify that $f^{-1}(x) = f(x)$.

Sketch the graph of:

$$f(x) = (2 - x)(x + 3)(x + 1)^2$$

Label the x -intercepts and y -intercept.



Question 30

1 mark

122

Which expression has a larger value?

$$\log_2 36 \text{ or } \log_3 80$$

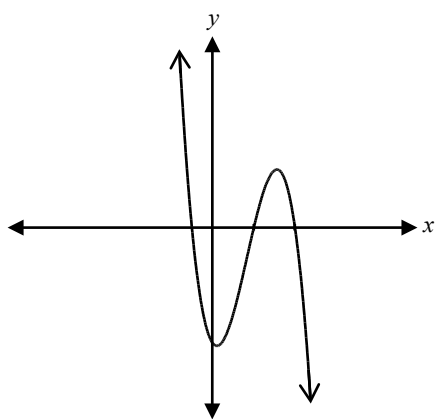
Justify your answer.

Question 31

1 mark

123

The graph below represents the equation $y = ax^3 + 6x^2 + 5x - 10$.



What must be true about the value of a ? Explain your reasoning.

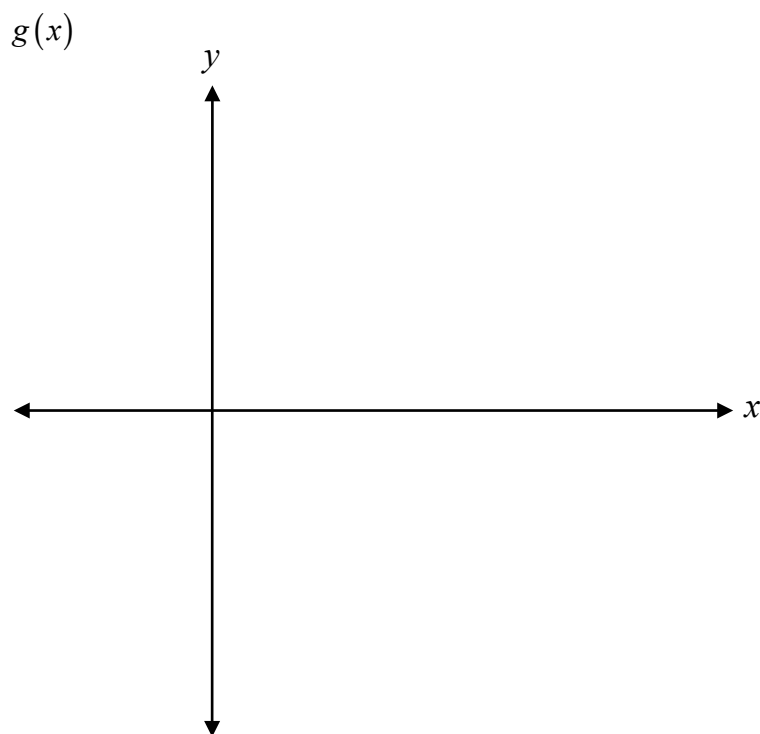
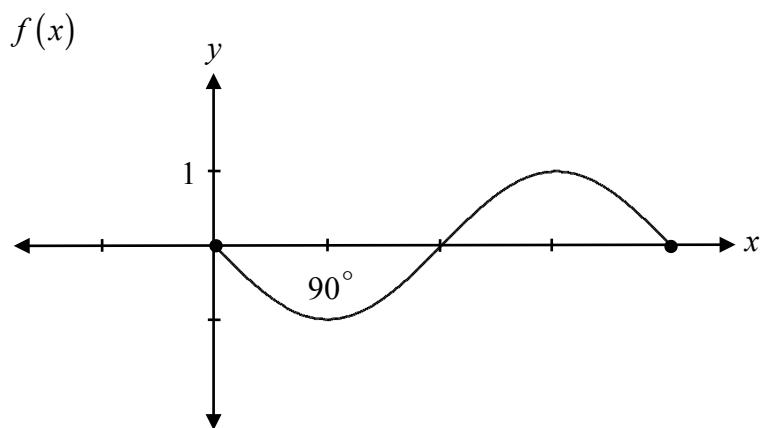
Question 32

2 marks

124

The terminal arm of an angle θ , in standard position, intersects the unit circle in Quadrant IV at a point $P\left(\frac{\sqrt{5}}{4}, y\right)$. Determine the value of $\sin \theta$.

Given the sinusoidal function $f(x)$ below, sketch the graph of $g(x) = |f(x)| - 1$.



Question 34**2 marks**

126

The graph of a rational function, $f(x)$, has a point of discontinuity when $x = 2$ and an asymptote when $x = 4$. Write a possible equation for $f(x)$.

Question 35**2 marks**

127

Given that $(x - 1)$ is one of the factors, express $x^3 - 57x + 56$ as a product of factors.

Question 36

2 marks

128

Give an example using values for A and B , in degrees or radians, to verify that $\cos(A + B) = \cos A + \cos B$ is **not** an identity.

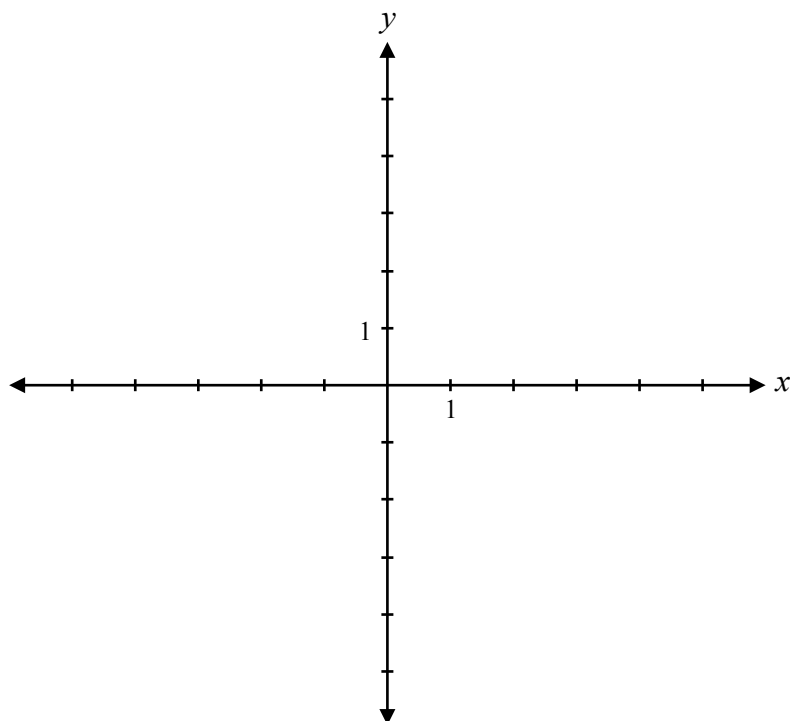
Left-Hand Side	Right-Hand Side

Question 37

3 marks

129

Sketch the graph of $y = \sqrt{x+1} - 2$ and verify that the value of the x -intercept is the same as the solution to the equation $\sqrt{x+1} - 2 = 0$.

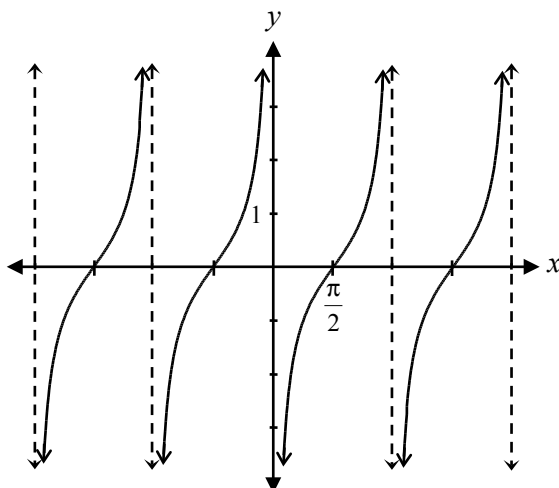


Question 38

1 mark

130

Mohamed is asked to sketch the graph of $y = \tan x$.
His graph is shown below.



Explain why his graph is incorrect.

Question 39

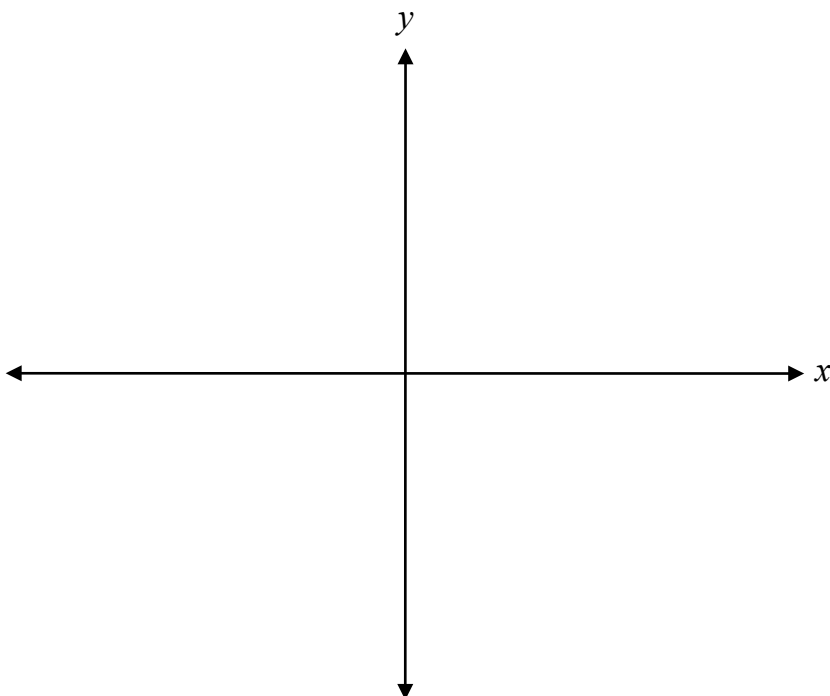
2 marks

131

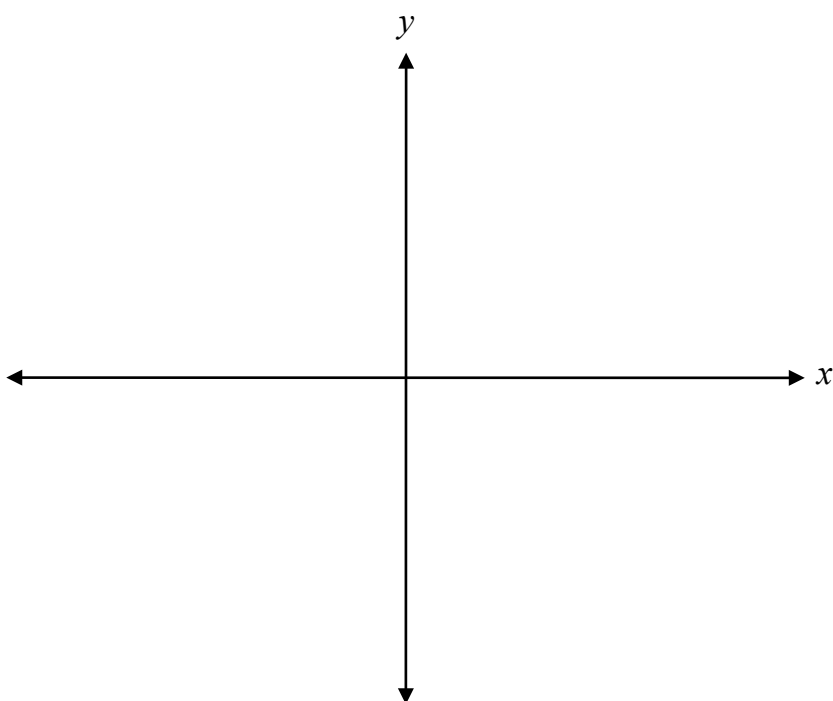
On the interval $0 \leq \theta < 2\pi$, identify the non-permissible values of θ for the trigonometric identity:

$$\tan \theta = \frac{1}{\cot \theta}$$

- a) Sketch the graph of $y = \ln(x)$.



- b) Sketch the graph of $y = -\ln(x - 2)$.



Question 41

2 marks

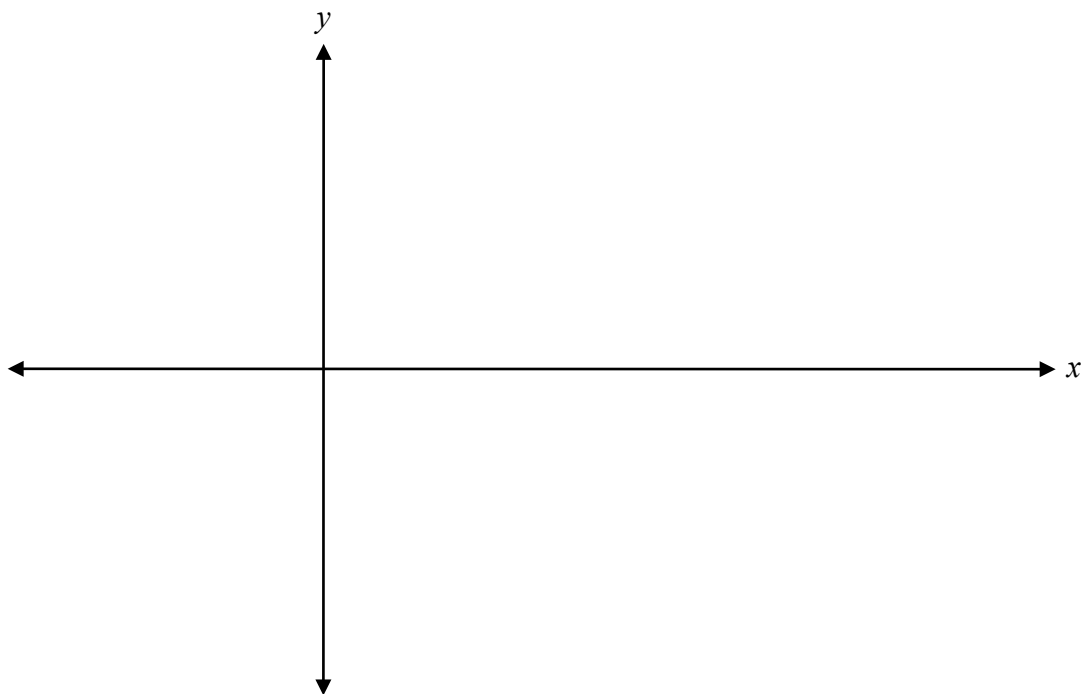
134

Given $f(x) = \sqrt{x - 2}$ and $g(x) = 3x$, write the equation for $h(x) = f(g(x))$.

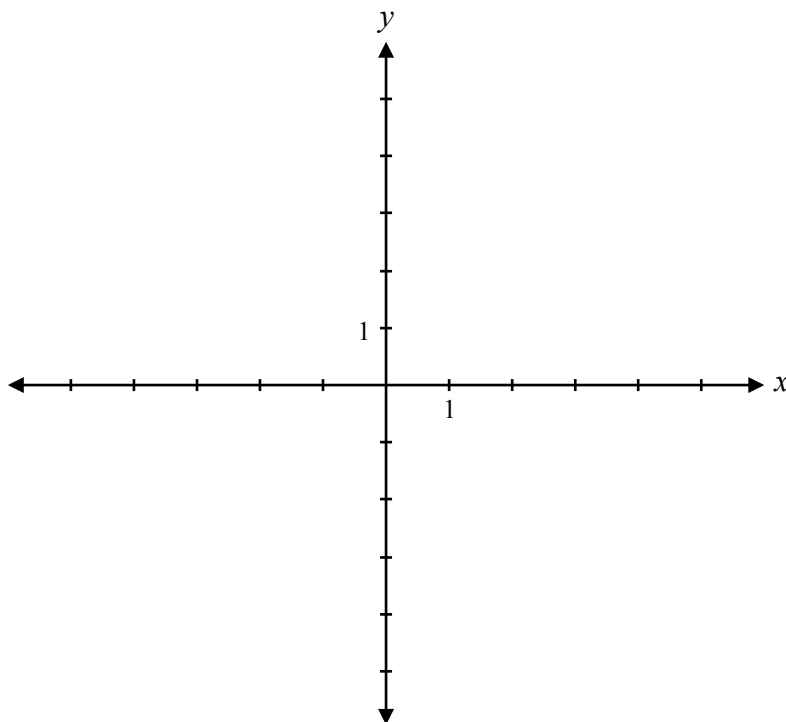
What are the restrictions on the domain of $h(x)$?

Explain your reasoning.

Sketch the graph of $y = 10\cos\left[\frac{\pi}{2}(x - 2)\right]$ over the interval $[0, 6]$.



Sketch the graph of the function $f(x) = \frac{x^2}{x^2 - x}$.



Question 44

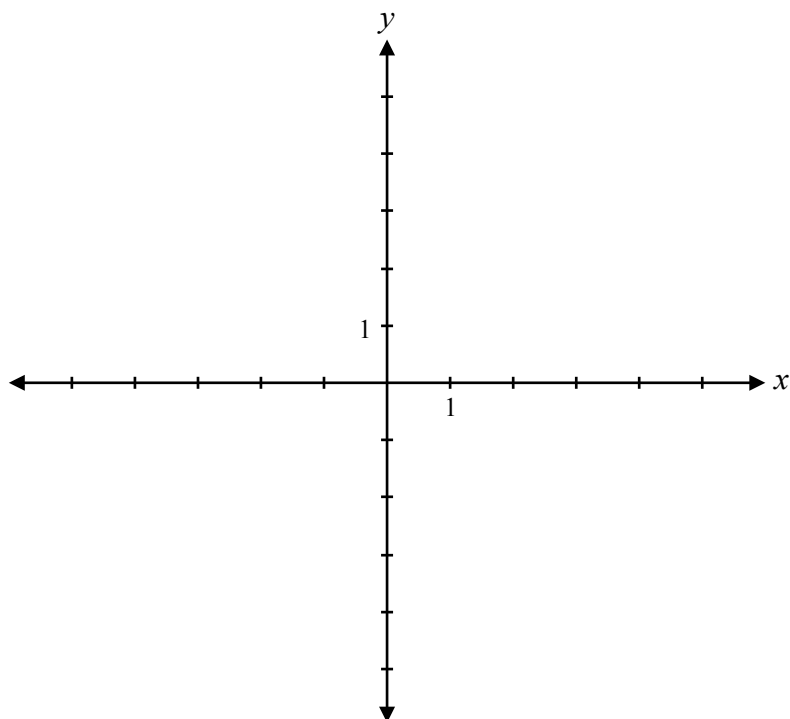
2 marks

137

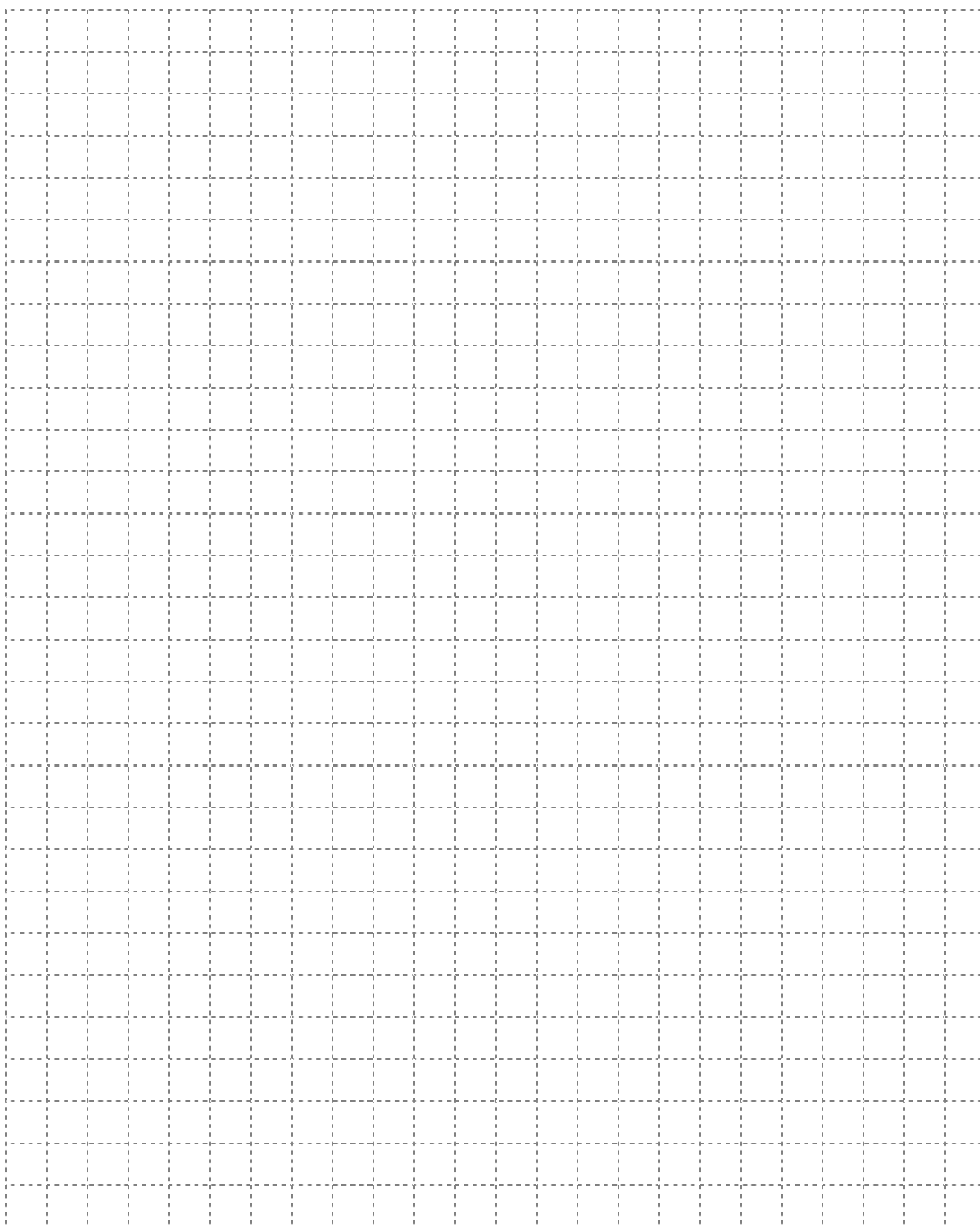
Is $(x - 3)$ a factor of $x^4 - x^3 - 3x^2 + x - 1$?

Justify your answer.

Given $f(x) = x - 1$ and $g(x) = x^2$, write the equation of $y = f(g(x))$ and sketch the graph.



No marks will be awarded for work done on this page.



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